

Emotion Recognition from Facial Images

Does Transfer Learning Improve Facial Emotion Classification Over a Baseline CNN?

Daniel Wei, Enogayin Imasuen, Suffian Haroon

Group 4 | CS1090B, Spring 2026

May 12, 2026

Abstract

We investigate whether ImageNet-pretrained transfer learning improves facial emotion classification over a custom convolutional neural network (CNN) trained from scratch on the FER-2013 benchmark. FER-2013 presents 35,887 low-resolution (48×48 grayscale) images across seven emotion categories with a severe class imbalance (16:1 ratio). Our experiments span four phases: a baseline CNN capacity study, a ResNet-50 transfer learning study with classical upscaling interpolation methods, a multi-model benchmark (VGG-16, ResNet-50, EfficientNet-B0), and a systematic ablation of augmentation, loss functions, hyperparameters, and fine-tuning depth. ResNet-50 with partial fine-tuning (last two blocks + head) achieves a best weighted F1 of 0.696 and roughly 70% accuracy, compared to 0.194 weighted F1 for the best baseline CNN. We find that model architecture is the dominant performance driver, that interpolation method is largely inconsequential (1–3% F1 difference), and that partial rather than full fine-tuning is essential on noisy, imbalanced datasets. Grad-CAM analysis confirms that transfer learning models attend to semantically meaningful facial regions, providing qualitative evidence of genuine emotion-related feature learning.

1 Background and Motivation

Automatic facial emotion recognition (FER) has wide-ranging applications in human-computer interaction, mental health screening, driver monitoring, and accessibility tools for individuals with autism spectrum conditions. Despite steady progress in deep learning, deployment in unconstrained environments remains difficult due to varying face angles, inconsistent lighting, partial occlusion, and low image resolution.

FER-2013, introduced by Goodfellow et al. (2013), is a standard benchmark of 35,887 low-resolution (48×48 grayscale) images collected via search engine queries — realistic but challenging. Our work addresses a practical question for any developer building an emotion-aware application: does the added complexity of transfer learning outperform a carefully tuned custom CNN, and what are the implications for computational budget and model interpretability?

A critical dimension of FER research is the intersection of class imbalance with demographic representation. FER-2013 was scraped without demographic controls, so underrepresented emotion classes (disgust at 1.5% versus happy at 25%) may map onto underrepresented racial or gender groups in the data. Failures on minority classes are therefore not neutral: if errors cluster by race or gender, the system operationalizes bias. Buolamwini and Gebru (2018) documented precisely this pattern in commercial facial analysis systems, finding significantly higher error rates for darker-skinned individuals and women. This is why we prioritize macro F1 and per-class recall alongside accuracy throughout our experiments.

2 Problem Statement

Research question: Does ImageNet-pretrained transfer learning improve facial emotion classification performance over a baseline CNN trained from scratch on a low-resolution, class-imbalanced dataset?

Our target output is a seven-class emotion label drawn from: angry, disgust, fear, happy, neutral, sad, and surprise. The unit of observation is a single 48×48 grayscale image. We refined our research question across milestones to emphasize robustness over raw accuracy: because class imbalance is severe and the dataset contains label noise, a model that achieves high overall accuracy by ignoring minority classes is not genuinely useful. We therefore report both weighted and macro F1 alongside accuracy, and we specifically track minority-class recall throughout.

3 Data and EDA

3.1 Dataset

FER-2013 contains 35,887 images split into 28,709 training and 7,178 test examples. Class counts are: happy (8,989), neutral (6,198), sad (6,077), angry (4,953), fear (5,121), surprise (4,002), and disgust (547). The 16:1 ratio between happy and disgust is the most extreme imbalance and strongly penalizes naive cross-entropy minimization.

Emotion	Train	Test	Total	% of dataset
Angry	3,995	958	4,953	13.8%
Disgust	436	111	547	1.5%
Fear	4,097	1,024	5,121	14.3%
Happy	7,215	1,774	8,989	25.0%
Neutral	4,965	1,233	6,198	17.3%
Sad	4,830	1,247	6,077	16.9%
Surprise	3,171	831	4,002	11.2%

Table 1: FER-2013 class distribution. Disgust is severely underrepresented at only 1.5% of the dataset.

3.2 EDA Findings

Three EDA analyses materially influenced our modeling decisions. First, per-class distributions of mean pixel intensity and pixel standard deviation are nearly indistinguishable across all seven emotion categories (Figure 1). This confirmed that raw intensity statistics carry little discriminative signal, and that models relying on global pixel features would fail—motivating spatial feature learning via convolution.

Second, mean FFT magnitude spectra across emotion classes are nearly identical in structure, all dominated by low-frequency components. Emotions are essentially indistinguishable in frequency space. This reinforced the need for architectures that capture spatial geometry (eye shape, mouth curvature, brow position) rather than texture frequency.

Third, visual inspection of the training set revealed a meaningful number of noisy samples: duplicate groups of up to 14 near-identical images across different emotion labels, placeholder images (warning triangles, missing-image icons), and images of non-face objects. Although for MS4 we reverted to the original unmodified FER-2013 train

split to ensure reproducibility, these quality issues motivated our use of data augmentation and robust loss functions rather than assuming clean labels.

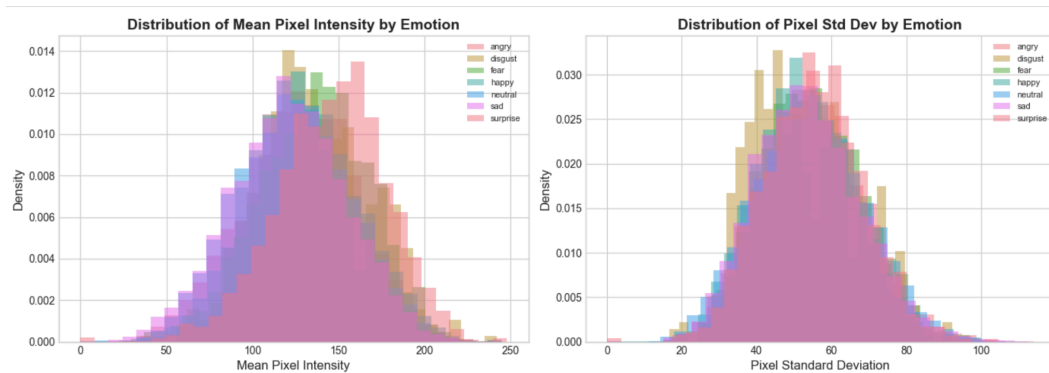


Figure 1: FER-2013 distribution of mean pixel intensity and pixel standard deviation spread

4 Methods and Models

4.1 Preprocessing Pipeline

All images are loaded as RGB (three-channel), normalized using ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]) for transfer learning models, or grayscale-normalized to mean 0.5 / std 0.5 for the baseline CNN. Because ImageNet-pretrained models expect 224×224 inputs, we upscale from 48×48 using classical interpolation. We compare four methods: Nearest Neighbor, Bilinear, Bicubic, and Lanczos. We use the FER-2013 canonical train/test split throughout (28,709 train, 7,178 test) without carving out a separate validation set; all reported metrics are on the held-out test set. This decision was made to maximize training data given the dataset's limited size and class imbalance, and to maintain comparability with prior work on this benchmark.

Our baseline augmentation strategy (applied to the training set only) uses random horizontal flipping ($p=0.5$), random rotation ($\pm 15^\circ$), and color jitter (brightness $\pm 20\%$, contrast $\pm 20\%$). This is the minimum set motivated by natural face variation. We then perform an ablation comparing five cumulative augmentation levels: None, Baseline (flip+rotation+color), +Scaling, +Translation, +RandomCrop, and +RandomErasing.

4.2 Baseline CNN Architecture

We designed four CNN architectures of increasing capacity to study the effect of model width and depth on FER performance. All share a common block pattern: Conv2d $\rightarrow$ BatchNorm2d $\rightarrow$ ReLU $\rightarrow$ MaxPool2d, applied multiple times. The models differ in channel widths, number of convolutional stages, fully connected layer depth, and dropout strength. Run1 (small, 2 conv stages, 32–64 channels) serves as an underfitting check; Run2 (baseline, 4 stages, 32–64–128–128) is the main reference; Run3 (deep, 4 stages, 32–64–128–256) increases abstraction; and Run4 (high-capacity, 4 stages, 64–128–256–512) is an overfitting stress test. All CNN models operate on grayscale 224×224 inputs and are trained from scratch.

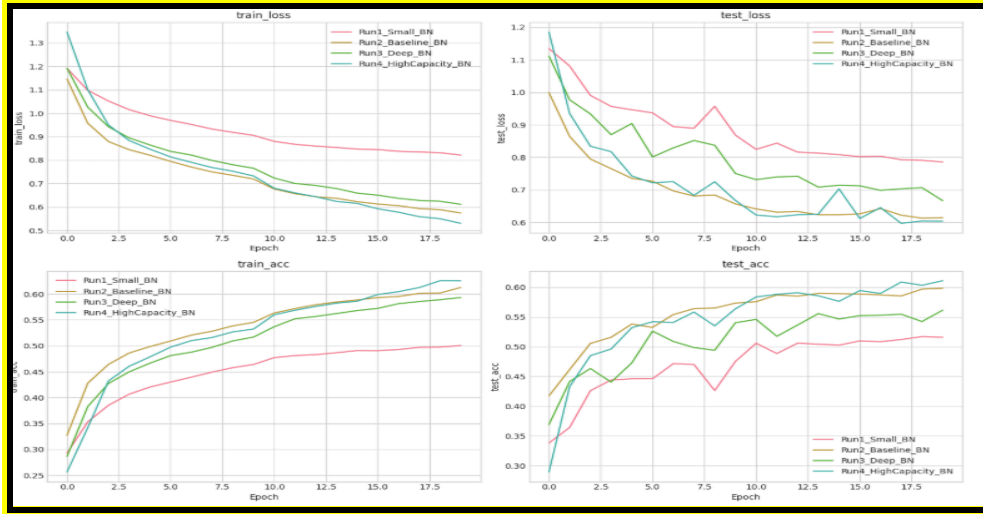


Figure 2: Training and Validation Curves for Baseline CNN Architectures with Batch Normalization

4.3 Transfer Learning Models

We evaluate three pretrained models from torchvision: ResNet-50 (25.6M parameters), VGG-16 (138M parameters), and EfficientNet-B0 (5.3M parameters), all pretrained on ImageNet-1K. We replace the final classification head with a seven-class linear layer. Training follows a two-stage protocol. Stage 1 (5 epochs) freezes the entire backbone and trains only the head using Adam ($\text{lr}=1\text{e-}3$). Stage 2 (15 epochs) unfreezes and fine-tunes the full network with a lower backbone learning rate ($\text{lr}=1\text{e-}4$) and cosine annealing schedule.

4.4 Class Imbalance Strategies

We test five strategies on ResNet-50: Standard Cross-Entropy, Weighted Cross-Entropy (class weights inversely proportional to frequency), Focal Loss ($\gamma=2$), Weighted Random Sampling (WRS) + Standard CE, and WRS + Weighted CE. Class weights are computed as sklearn's 'balanced' mode on the full training set.

4.5 Fine-Tuning Depth Study

We systematically vary how much of the ResNet-50 and VGG-16 backbone is unfrozen: Head Only (classification layer only), Last Block + Head, Last 2 Blocks + Head, and Full Fine-Tune. All four strategies are trained for 20 epochs with cosine LR annealing. This ablation tests whether full fine-tuning is beneficial or harmful given FER-2013's noisy labels and limited size.

4.6 Grad-CAM Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM, Selvaraju et al. 2017) to both the best CNN and the best ResNet-50. Grad-CAM computes the gradient of the class score with respect to the final convolutional feature map, produces a per-spatial-location importance weight, and overlays a heatmap on the original image. This provides a qualitative check on whether models are attending to facial geometry or confounding background patterns.

5 Results and Interpretation

5.1 CNN Capacity Study (MS3 Baseline)

All four CNN architectures were trained on 48×48 grayscale images using Focal Loss and Adam (lr=1e-3, wd=1e-4). Results are shown in Table 2. Increasing capacity consistently improved performance up to Run4, but the gap between accuracy and Macro F1 across all runs reveals a persistent bias toward majority classes. In the best CNN configuration (Run4_HighCapacity_BN), the disgust class achieved near-zero recall, confirming that capacity alone cannot overcome severe class imbalance. These CNN results constitute our MS3 baseline. The best MS3 CNN achieved 0.611 accuracy and 0.511 Macro F1, while our MS3 frozen ResNet-50 reached only 0.445 accuracy and 0.356 Macro F1 — generic ImageNet features without facial adaptation proved insufficient. MS4 directly addressed both limitations: our final ResNet-50 with partial fine-tuning achieves 0.70 accuracy and 0.696 weighted F1, a 14.5 percentage point gain over the MS3 CNN and a 3.6× improvement in weighted F1 over the MS3 transfer learning baseline.

Model	Accuracy	Macro F1	Weighted F1	Precision (M)	Recall (M)
Run1_Small_BN	0.516	0.429	0.512	0.424	0.436
Run2_Baseline_BN	0.598	0.489	0.579	0.492	0.499
Run3_Deep_BN	0.561	0.463	0.551	0.522	0.457
Run4_HighCapacity_BN	0.611	0.511	0.598	0.644	0.520

Table 2: CNN capacity study results (MS3 baseline). Run4 performs best but the Macro F1 gap reveals class imbalance bias.

5.2 CNN Ablation Study

We applied the Run4 high-capacity architecture to a systematic ablation varying loss function, optimizer, learning rate, and weight decay. Key results: the Weighted Sampler + standard CE configuration achieved the best Macro F1 of 0.535 and best macro recall of 0.567, outperforming even the baseline by providing more balanced gradient updates. Standard cross-entropy baseline accuracy was 0.592, while Weighted CE reached 0.587, and Weighted Sampler CE reached 0.582 accuracy but 0.535 Macro F1 — a substantially better class-balanced result. Lower learning rates (1e-5) dramatically degraded performance, and SGD was consistently inferior to Adam on this dataset. The class-wise F1 breakdown reveals that happy (F1 ~0.80–0.83) and surprise (~0.70–0.74) are reliably detected, while disgust (F1 0.00–0.27) and fear (0.15–0.35) remain the hardest classes. The Weighted Sampler configuration provides the most uplift for disgust, raising it from near-zero to approximately 0.27 F1.

5.3 Transfer Learning: Interpolation and Architecture

The Phase 1 vs Phase 2 comparison is striking. The best BaselineCNN achieved weighted F1 of only 0.194 on 224×224 upscaled inputs, while ResNet-50 transfer learning reached 0.696 weighted F1 — a 3.6× improvement. This directly and definitively answers our research question: transfer learning substantially outperforms training from scratch on this dataset.

Phase	Model	Upscaling	Val Accuracy	Weighted F1
Phase 1 — Baseline CNN	BaselineCNN	Best (Nearest N.)	~0.31	0.194
Phase 2 — ResNet-50 TL	ResNet-50	Lanczos	~0.70	0.696
Phase 3 — Multi-Model	VGG-16	Bicubic	~0.57	0.574

Phase	Model	Upscaling	Val Accuracy	Weighted F1
Phase 3 — Multi-Model	ResNet-50	Bilinear	~0.58	0.580
Phase 3 — Multi-Model	EfficientNet-B0	Bilinear	~0.55	0.553
Phase 3 — Multi-Model	BaselineCNN	Bicubic	~0.25	0.046

Table 3: Phase 1 vs Phase 2 vs Phase 3 best results by model and upscaling method. Transfer learning models cluster tightly between F1 0.55–0.70.

The interpolation method had minimal impact on transfer learning performance: across all four upscaling strategies, the weighted F1 range for ResNet-50 in Phase 3 was only 0.553–0.580, a 2.7% spread. Bicubic and Lanczos performed marginally better for ResNet-50, while differences for VGG-16 and EfficientNet-B0 were similarly small. The implication is that practitioners need not invest in learned super-resolution preprocessing when using pretrained models.

5.4 Augmentation Ablation

For ResNet-50, the baseline augmentation (flip + rotation + color jitter) already captures most of the benefit, reaching weighted F1 of 0.577 from a no-augmentation baseline of 0.544. Adding scaling raised this to 0.584. Additional geometric transforms (translation, random crop) and random erasing provided marginal gains (0.576, 0.574, 0.579 respectively), showing diminishing returns. For the baseline CNN, augmentation results were largely noise around very low baseline values (0.000–0.169 weighted F1), confirming that augmentation cannot compensate for insufficient representational capacity.

5.5 Class Imbalance — Loss and Sampling (ResNet-50)

Weighted CE achieved the best result at 0.592 weighted F1, narrowly outperforming standard CE (0.583), Focal Loss (0.581), WRS + Standard CE (0.576), and WRS + Weighted CE (0.577). The margins are small (within 1.6%), suggesting ResNet-50's pretrained representations are sufficiently robust that loss function choice is secondary once the backbone is adapted. Notably, Focal Loss did not emerge as a clear winner in the transfer learning setting, in contrast to the CNN ablation where Weighted Sampler provided the largest minority-class boost.

5.6 Hyperparameter Sensitivity (ResNet-50)

Adam ($\text{lr}=1\text{e-}3$) outperformed SGD with momentum (0.578 vs 0.460 weighted F1), confirming Adam as the appropriate optimizer. Learning rate sensitivity was high: reducing lr to $1\text{e-}4$ dropped F1 to 0.544, and $1\text{e-}5$ dropped further to 0.314. Weight decay had negligible effect (0.578 at $\text{wd}=1\text{e-}4$, 0.579 at $\text{wd}=0$, 0.582 at $\text{wd}=1\text{e-}3$), suggesting the pretrained backbone already provides sufficient implicit regularization.

5.7 Fine-Tuning Depth Study

This is among the most practically important findings. For ResNet-50, partial fine-tuning of the last two blocks plus head achieved 0.579 weighted F1, substantially outperforming head-only training (0.361) and also outperforming last-block + head (0.535). Critically, full fine-tuning catastrophically degraded performance to just 0.126 weighted F1 — a near-total loss of pretrained representations when all parameters are simultaneously updated on noisy FER-2013 labels. The same pattern held for VGG-16: last 2 blocks + head achieved 0.571, while full fine-tuning collapsed to 0.147.

Strategy	ResNet-50 F1	VGG-16 F1
Head Only	0.361	0.339
Last Block + Head	0.535	0.554
Last 2 Blocks + Head	0.579	0.571
Full Fine-Tune	0.126	0.147

Table 4: Fine-tuning depth study. Full fine-tuning catastrophically degrades both models; last 2 blocks + head is optimal.

5.8 Grad-CAM Interpretation

Grad-CAM heatmaps from the best baseline CNN show reasonable but diffuse attention, often spreading across the central face region with some focus on the nose and mouth area. Correctly classified samples (angry, happy, neutral) showed tighter activation on diagnostic regions. Misclassified samples (disgust predicted as surprise, sad predicted as fear) showed the CNN attending to incorrect or peripheral regions.

ResNet-50 Grad-CAM heatmaps demonstrated substantially more semantically coherent attention. Happy predictions concentrated on the mouth (smile region), angry predictions activated strongly on the brow-eye region, and surprise activations centered on the widened eyes. This qualitative difference provides evidence that transfer learning genuinely produces more meaningful facial feature representations, not just higher-accuracy pattern matching.

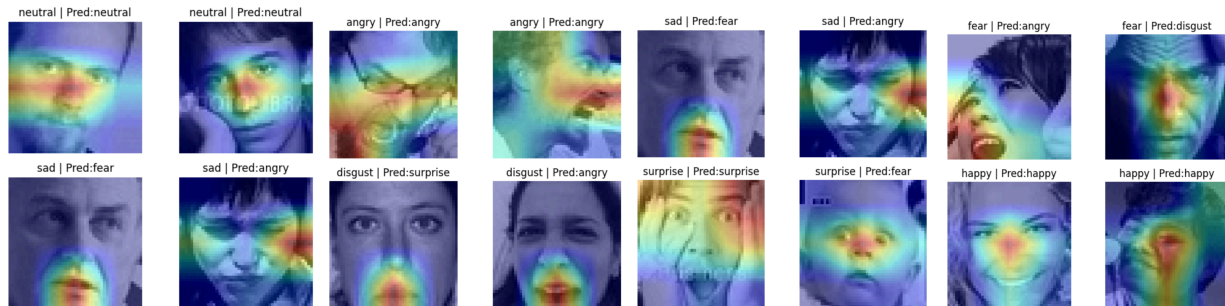


Figure 3: TCNN Grad-CAM that has correct predictions (angry, happy, neutral) + misclassifications (disgust $\rightarrow$ surprise, fear $\rightarrow$ angry, etc.)

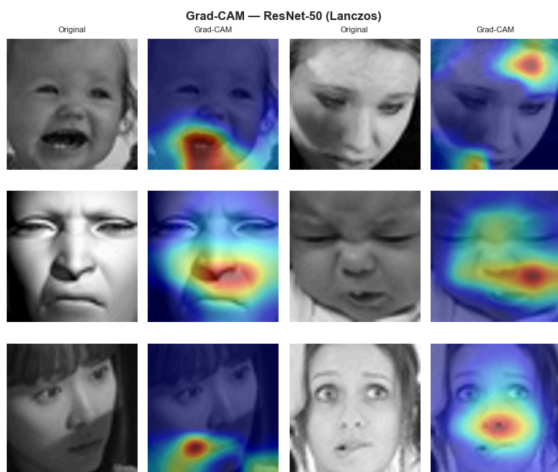


Figure 4: Grad-CAM heatmaps from ResNet-50 (Lanczos). Activations concentrate on diagnostic regions — mouth for happy, brow for angry — confirming genuine emotion-relevant feature learning.

6 Conclusions and Discussion

Our central finding is clear: transfer learning from ImageNet substantially outperforms training a custom CNN from scratch on FER-2013. ResNet-50 with the last two blocks fine-tuned achieves weighted F1 of 0.696 ($3.6\times$ better than the best CNN- 0.194) and Grad-CAM evidence confirms qualitatively superior feature learning. This answers our research question affirmatively and with a large margin.

Several secondary findings are worth emphasizing. First, the interpolation method is not a meaningful design decision for practitioners: all four classical methods (nearest neighbor, bilinear, bicubic, Lanczos) produce within 3% of each other on any fixed model, and bicubic or Lanczos offer marginal advantages at no cost. Second, partial fine-tuning is essential, not optional. Full fine-tuning is catastrophic on FER-2013, collapsing weighted F1 from 0.579 to 0.126 for ResNet-50. This is consistent with established transfer learning theory: catastrophic forgetting occurs when the fine-tuning dataset is much smaller and noisier than the pretraining corpus. Third, augmentation is most valuable in moderate amounts— flip, rotation, and color jitter capture most of the available benefit, and adding random erasing or aggressive cropping provides diminishing returns.

What we cannot claim: our results are limited to a single benchmark under a fixed set of architectures. FER-2013 is widely considered to have noisy labels, and some of our findings (e.g., the superiority of Adam over SGD) may not generalize across datasets with different noise profiles. We also do not evaluate inference latency or model size, which matter for deployment on edge devices. Our experiments did not include Vision Transformer architectures or more recent efficient backbones, which may behave differently. The most fundamental limitation of our approach is the supervision signal: FER-2013's hard one-hot labels poorly reflect the continuous, ambiguous nature of human emotion — fear and surprise share overlapping facial configurations, and a single image can plausibly carry multiple labels. Soft re-labeling from annotator agreement distributions, or joint valence/arousal prediction alongside discrete classes, would better match this structure and likely reduce the disgust/fear confusions we observed. A second priority is demographic robustness: our Grad-CAM analysis confirms correct average attention, but average behavior conceals whether that reliability holds across skin tones and genders. Fairness-aware training objectives and demographically stratified evaluation sets should be built in from the outset, not treated as a post-hoc audit. These two directions with better supervision and fairer evaluation, are more likely to produce meaningful real-world gains than architectural scaling alone.

7 Broader Impact

Emotion recognition from facial images raises significant ethical considerations that any deployment must address. On the positive side, accurate FER can benefit people who have difficulty reading social cues (e.g., in assistive technology for autism spectrum disorder), enable real-time mental health monitoring applications, and improve driver safety systems. These applications could reduce harm and improve quality of life.

However, the potential harms are substantial. Emotion inference from faces is not reliable across demographic groups — published studies have documented higher error rates for darker-skinned faces and for women in several commercial FER systems. FER-2013 itself was collected via search engine queries and may not represent demographic diversity. A model trained on this data could perpetuate or amplify these disparities if deployed in high-stakes settings such as hiring, policing, or mental health triage. Additionally, FER systems could be used for covert emotional surveillance without consent, with implications for privacy and civil liberties.

We recommend that any production deployment of FER systems include: (1) demographic bias audits before release, (2) explicit consent from people being monitored, (3) human oversight for consequential decisions, and (4) opt-out mechanisms. Our work is a research study on a public benchmark and should not be interpreted as a production-ready system.

References

- [1] Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Chumerin, N., & Bengio, Y. (2013). Challenges in representation learning: A report on three machine learning contests. *ICML 2013 Workshop on Challenges in Representation Learning*, arXiv:1307.0414.
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep residual learning for image recognition*. In Proceedings of CVPR.
- [3] Simonyan, K., & Zisserman, A. (2014). *Very deep convolutional networks for large-scale image recognition*. arXiv:1409.1556.
- [4] Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking model scaling for convolutional neural networks*. In Proceedings of ICML.
- [5] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: *Visual explanations from deep networks via gradient-based localization*. In Proceedings of ICCV.
- [6] Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). *Focal loss for dense object detection*. In Proceedings of ICCV.
- [7] FER-2013 Dataset. Available at: <https://www.kaggle.com/datasets/msambare/fer2013>
- [8] PyTorch / TorchVision. (2023). torchvision.models. <https://pytorch.org/vision/stable/models.html>
- [9] Buolamwini, J., & Gebru, T. (2018). *Gender shades: Intersectional accuracy disparities in commercial gender classification*. In Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT* 2018), PMLR 81, pp. 77–91.
- [10] AI Assistance: Claude (Anthropic) was used for report editing/grammar checks. All experiments, analysis, code, and conclusions are the work of the authors.